

EnglishPod365 0282

Global View - Physics

GENERATED SUPPLEMENT

The original host-script PDF was not available in the public archive. This file is generated from the local complete lesson transcript and is provided as a usable transcript supplement; it is not an original publisher PDF.

Complete lesson transcript

Hello everyone and welcome to EnglishPod. My name is Marco.

My name is Catherine and today we have an advanced level lesson for you. This one's a bit difficult. It's all about science.

That's right. Today we're going to talk about physics, specifically about, well you know, things like Stephen Hawking and wormholes, black holes and general theory of relativity and all that good stuff.

Right. So we're in class. We're learning about physics today and you're going to hear a lot of terms. Don't worry if you're not really sure what they mean. We'll be back in a moment to talk about them.

Good morning, everybody. Welcome to Physics 101. My name is Ed Brown and I will be your professor for the semester.

Since today is our first day, I wanted to give you an overview of what this course will look like, how you will be graded and what we will cover this semester.

Will we be focusing more on theoretical physics or experimental physics, professor?

This is an introductory course and my aim is to give you a broad overview of the field of physics.

The term physics encompasses many different areas of research and study.

And I hope this course will provide you with a conceptual understanding of physics, which will prove useful whether or not you choose to further your study in the field.

We will begin the course by looking at the fundamental concepts of physics.

Then by the middle of the semester, we will begin exploring the more theoretical side of physics.

It is essential that you first have a firm grasp of the fundamentals so that you can better understand the theoretical concepts when we get to them.

Will we learn about black holes, wormholes and string theory?

We will learn about the general theory of relativity, including black holes.

We will also explore the developing theories in quantum mechanics, such as string theory.

We will discuss some hypothetical features of space-time, like wormholes.

We will also explore some of the more influential developments in the field of thermodynamics, electromagnetism and nuclear physics, all of which have had significant impacts on modern life.

Now I am going to have the TAs pass out the syllabus for this class, so you can see how this course will be graded.

Oh man, looks like this isn't going to be the easy A I thought it would be.

Alright, we're back. So obviously we are in a class, as you said, and the professor is pretty much introducing the course and what's going to happen during the semester.

That's right. So he's talking about the different kinds of physics or the different elements of physics that will be a part of this course.

And there's a very eager student, he's very excited and he wants to know, are these interesting things going to be a part of the course?

And so when we start out, we talked about these two different fields of physics and I think maybe it would be helpful to start there with these two branches of this field.

That's right. He mentioned, this boy Matt said, if they're going to be focusing more on theoretical physics or experimental physics.

Now, theoretical physics and experimental physics are what we call two different branches of physics. That means they're both physics.

But people who do them have different methods and maybe focus on different things.

Theoretical comes from the word theory. Right.

So I have a theory about something and I'm going to think about whether or not it's possible and maybe there's some math involved.

But with experimental physics, the key word here is experiment, experimental.

That means we actually build experiments to test our theories.

That's right. So I guess theoretical physics, we can just have theories about how things happen or explain them, but we can't really prove them.

We can just have theories around them. Maybe we can try and prove them mathematically.

But experimental physics, you can actually, as you said, create an experiment to prove it or not.

That's right. And so those are the two main branches that the young man asks about.

But the professor says, you know, listen, we're going to be talking about the fundamental concepts of physics.

And so fundamental concepts are things that you often hear a professor talk about in your introductory class.

So introduction to mathematics or introduction to physics.

So fundamental concepts are the basics.

That's right. So the most important, but also the ones at the very bottom, the foundation.

Because you can learn about more difficult or more complicated things later.

But first, you have to have a foundation. So those are the fundamental concepts.

Right. And actually, after the professor said that, this other, this guy, Matt, went on to ask if they were going to learn about black holes, wormholes and string theory.

OK, so three very important parts of physics, but very, very complicated things.

The first two are related, right? So we've got astrophysics.

Astro means star. So astrophysics is the physics of things, bodies in space, planets, stars, things like that.

That's right. And so two concepts that we often talk about in astrophysics are black holes and wormholes.

That's right. So I think many of you have already heard this word, black hole.

The concept of it is basically a huge hole in space that just sucks up everything.

The gravitational force. So the pull of gravity is so strong that nothing can resist it.

Everything that comes close to a black hole, it's like in a drain.

Everything that comes close to it gets sucked in. No light can escape it.

That's right. And the other concept there is a wormhole.

So it's basically a sort of bending of space and time in which you can travel from one point to another a lot faster than going in a straight line.

Or from one time to another. So this is a very, very popular concept in science fiction movies and books.

For example, Star Trek, the most recent movie, the new movie Star Trek, there was a wormhole.

So one ship goes from the future and it comes into the past.

That's right. So it has to travel through this wormhole.

That's right. And actually, yeah, you say science fiction and obviously it's hard to prove it, but this is one of the concepts that physicists think they can prove someday.

Someday. And this is why it's a part of theoretical physics.

That's right. And what about this other concept, string theory?

Okay, string theory is a theory about the way the world works, basically.

And it is a theory that actually competes with Einstein's view of the way the world works, right?

And string theory is about basically strings that are vibrating and that's the way the entire universe is kind of built.

Yeah, so basically at the subatomic level, usually you're supposed to have an atom,

but also supposedly these atoms are made up of strings that depending on the frequency at which they vibrate,

they can constitute different elements or different parts.

Different kinds of matter, right.

Exactly.

And so we're talking about the very, very, very, very, very, very, very smallest building blocks of the things in the world,

atoms and even quarks and gluons and those things.

Exactly. So it's actually one of these new things that, well, people are talking about string theory.

It's kind of this new way of seeing the world and explaining the unexplainable

because there are so many things still in the world that we can't explain.

And people argue about this because obviously if you can't prove it, it's hard to say what is right and what is wrong.

Exactly. But actually one thing that was proven was Einstein's general theory of relativity.

So this is something we all learned back in high school or in college.

That's right. So basically the general theory of relativity is this idea that space and time are not fixed.

They're relative to each other.

So an example my teacher used to use is that when you travel very, very, very fast, imagine going around a jet around the world counterclockwise, time slows down the faster you move.

Okay, so that proves that time and space are not fixed.

That's right. Actually, the general theory of relativity and specifically this aspect of speed and time was proven with the example that you give with two atomic clocks.

You kept one on the ground, like, I don't know, in your house,

and you put one in an airplane that was traveling faster than the speed of sound.

And so after maybe an hour, the plane would land and the clocks would have not the same time anymore,

but they would be one, the clock that was in the airplane traveling faster would be maybe half a second off

or something very, very small. But you would see that time moves slower because your speed is faster.

All right. So that's the general theory. That's a part of the general theory of relativity.

It's a very important theory in the history of the world, but also in the history of physics.

Another thing, though, that the professor mentioned that is very important in physics these days is quantum mechanics.

Yes, this is actually a very interesting field of physics, but very hard to understand.

In my opinion, for me, it's very, very hard.

I find it very complex, too. Well, mechanics are the way things work, right?

And so quantum is at the atomic and subatomic level.

And so you're talking about basically the building blocks of the things in the universe.

And the people who are studying this or working on this, quantum mechanics, they don't like the Einsteinians.

They don't like the people who follow Einstein because these two theories, they don't actually go together very well.

You either have to believe one or you have to believe the other.

Well, the problem is that when you're talking at the macro level, the way things work and the way things move, like for example, you and me,

of course, there are formulas and there are everything that explains and that proves that things are the way they are.

But the problem is that in quantum mechanics at the subatomic level, it's so different.

Like, for example, electronics, the reason why we're able to have microprocessors and electronics is because,

for example, atoms can be in two places at the same time, which is impossible in our world.

It's like saying, you know, Catherine can be in two places at the exact same time.

And there's also this problem of observation that I've read about.

So there are some kinds of particles like electrons, for example, that if you're observing them, sometimes appear as waves.

But if you're observing them, sometimes appear to have mass.

Exactly like light.

Yeah. So you're talking about these things that seem to have a whole different set of rules from the rules that we have here walking around and eating food and going to work.

And so I think that's what's fascinating about physics, that, you know, you can break things down to the very smallest parts and they seem very different.

Or you can think about the stars and the moon and, you know, these planets and they have their own rules, too.

And so, you know, that's why physics is such a very, I think it's, it's, it's frustrating.

It's frustrating and it's exciting at the same time.

That's right. And actually some of the other fields that the professor said they were going to explore are the fields of thermodynamics, electromagnetism and even nuclear physics.

So very important branches, I think, of physics as well.

So when you're studying physics or science, you will definitely hear about the laws of thermodynamics.

Thermo means heat and dynamics means, you know, changing, changing state.

And electromagnetism, of course, is the combination of two words, electricity and magnetism.

You know, like magnets, you have the plus end and the minus end.

Right.

They attract or they repel.

Then finally, nuclear physics.

Nuclear physics is very important because it's, again, about the, about atoms and breaking down atoms and finding out what their parts are.

So you might have heard about the Hadron Super Collider.

Right.

This is like a big, it's like a big racetrack for atoms and they basically spin these atoms around the racetrack going faster and faster.

And then they put objects in their way so that they break.

They crash.

They crash.

And they study what the pieces look like.

And what happens and how much energy is released.

It's all very interesting and it seems like things that are out of science fiction because it's not something mainstream that you hear about in the news very often.

Or that sometimes we understand, like obviously we all hear about the gossip magazines and who's dating who and all this other type of news.

But very rarely do we really pay attention to this type of information or news because it is a little bit complex to understand.

And it seems that, I think scientists are not the best writers for average people like you and me.

They write in their own language almost.

Oftentimes that's the case.

But I have to say I recently read Bill Bryson's book, A Short History of Nearly Everything, and it talks about the history of the world and us and the animals and plants in it.

And he talks about physics especially.

And I think that if you're interested in that, I don't know, I mean if you're an advanced English learner, it might be challenging.

It was challenging for me.

But it's definitely a good book to read to learn about some of these things.

Yeah, it's actually a very good idea.

And I don't know, many of us, I think many of us as well growing up, maybe we had a hard time with some of these exact sciences like chemistry or physics.

So we obviously find it hard to be interested in it.

But you know it's very different because it's not math and formulas.

You don't really have to know how to derive the formula or anything.

You just kind of have to understand the theory.

And it's very, very interesting I must say.

It is and it's fun because you can talk about, it's like philosophy sometimes, you can talk about if this were the case, then what would happen.

These hypothetical situations and I think that's maybe more in line with theoretical physics.

But it's really interesting to think about and to let your mind go into these very strange places.

That's right.

So actually we are curious to know if you believe that maybe someday we will be able to travel in time.

Or do you believe that... Live on other planets or we'll have space travel as a part of our holiday making.

Yeah, they supposedly already are arranging tourism to the moon, right?

Yeah, they are. So I wonder if one day like in the fifth element we're going to have these cruises to space.

Yeah, that would be fun.

So let us know what your opinions are about this topic and also if you have any suggestions or questions, you can always visit us on our website englishpod.com.

We hope to see you there and until next time.

Bye.

Music.